

JULIA BUSH

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Here's my **portfolio website**, which I encourage you to check out: juliabush.dev

Frontend-focused developer and IB Diploma graduate with experience building full-stack web applications. Currently learning and building projects with React, Next.js, and interactive 3D web technologies.

EXPERIENCE

[Asteroids](#) - Full Stack App ([Github](#))

- Built a headless **Pygame** server, using **Python**, with an async 60 FPS loop and a low-latency **WebSocket** protocol for real-time state updates, collision detection, and input handling.
- Designed a responsive frontend with semantic **HTML** and **CSS** best practices (**BEM notation**, **root variables**), supporting real-time **multi-player** gameplay via client-side WebSockets.
- Automated **AWS backend** deployments via **GitHub Actions**, ensuring continuous delivery and persistence of multiplayer game state.

[3D Pokeball Catcher & Pokedex](#) - Full Stack App ([Github](#))

- Built an interactive Pokedex using **React**, **Typescript** and **Three.js** with a fully animated 3D Pokeball and probability-based catch logic tied to Pokemon rarity.
- Designed a **Node.js + Express** backend with custom **REST** endpoints and command-based game logic, managing in-memory game state and PokeAPI integration with caching, and **Vitest**-based unit testing.
- Deployed a containerized backend on **AWS EC2** with **Docker** and **Caddy**, as a reverse proxy, and a Vite-powered frontend on **Vercel**, establishing a production ready-full-stack deployment pipeline.

[Tarot AI](#) - Web App ([Github](#))

- Built an interactive Tarot reading app in **Next.js (App Router)** with **React + TypeScript**, streaming AI responses from the **OpenAI API** for real-time output.
- Implemented incremental rendering using a custom **React typewriter hook**, consuming streamed tokens and updating the user interface as data arrives.
- Developed a reusable, animated UI with **Tailwind CSS** and **Framer Motion**, including modal-based results, dynamic Tarot card visualization, and prompt logic driven by selected cards.

[Blog Aggregator](#) - CLI Backend Tool ([Github](#))

- Engineered a **middleware**-based CLI framework in **TypeScript** enabling modular command registration and execution, eliminating repetitive logic across **10+** commands.
- Modeled a normalized **PostgreSQL** schema using **Drizzle ORM**, implementing foreign key constraints, cascading deletes, and deduplication logic for consistent relational integrity.
- Developed a background feed aggregation loop with **Node.js** async timers and **XML** parsing for continuous **RSS** ingestion and post updates.

[Static Site Generator](#) - CLI Backend Tool ([Github](#))

- Built a static site generator in **Python**, implementing a custom Markdown parser, **DOM**-based **HTML** node system, and recursive directory traversal to generate HTML pages from .md files.
- Implemented a DOM-style node system (ParentNode, LeafNode, HTMLNode) to construct and render nested HTML structures dynamically in Python.
- Integrated **Pytest** and **GitHub Actions CI/CD** for automated test execution and coverage reporting (**~79% coverage**), applying test-driven development for all parsing and rendering logic.

[The Odin Project](#) - ([Github](#))

- Improved avatar hover behavior and copy-to-clipboard feedback after a Tailwind v4 upgrade, enhancing UI consistency for a platform serving over **1.7 million users**.
- Collaborated with contributors through **GitHub PRs**, incorporating feedback and refining code

[Lolnames](#) - ([Github](#))

- Improved mobile styling by refining **Tailwind CSS** classes for responsive spacing and layout, and component scaling.
- Implemented full light/dark theme support with **next-themes**, integrating browser preference detection.